Wang Jiawei

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https://drive.google.com/file/d/169v6_2eP32dJcDycq_qYqZRLbRMAP9gi/view?usp=sharing

EDUCATION

B.Eng (Computer Engineering) 08/2024 – Expected 07/2028 | Singapore
National University of Singapore
- Second Major in Innovation and Design
- GPA: 4.39/5

Nanyang Junior College 01/2020 – 12/2021 | Singapore
GCE A-Level: AAB/A

SKILLS

Programming Languages: C/C++, Java, Python, Verilog (HDL)
Embedded Systems: Arduino, Raspberry Pi, STM32, ESP32, Communication Protocols (SPI, I2C, UART)
Hardware Design: FPGA Design (Vivado), RTL Design, PCB Design (Fusion 360), Microsoldering
Tools & Software: Git/GitHub, Keil uVision 5, Xilinx Vivado, VS Code, Linux/Unix

PROFESSIONAL EXPERIENCE

Research Assistant 08/2025 – Present | Singapore
iHealthtech, National University of Singapore
Undergraduate Research Opportunities Program(UROP)
- Designed and prototyped a wireless, multi-site wearable sensor network for heat stress assessment.
- Custom PCB design (Fusion 360 Electronics), microsoldering, and circuit assembly of miniaturized sensor modules.
- Gained experience in embedded systems and wireless sensor integration.

PROJECTS

Smart Wearable Cuffless Blood Pressure Monitor (iDP) 01/2026 – Present
- Developing a cuffless blood pressure monitoring wearable to address limitations of intermittent, cuff-based measurement in continuous patient monitoring
- Targeting hypertension patients and remote healthcare use cases, enabling non-invasive, real-time tracking outside clinical settings
- Designing embedded system using PPG sensing with signal processing pipeline for physiological feature extraction and BP estimation
- Collaborating with industry partner Alexandra Hospital (AH) to align prototype with clinical requirements, usability, and validation standards
- Evaluating trade-offs between accuracy, wearability, and power consumption in wearable design

OPTIMEAL: AI-Powered Meal Planning Web App (Orbital) 05/2025 – 08/2025
- Built AI-powered meal planning web app using React (Next.js), Firebase, Node.js, and OpenAI API
- Implemented authentication, meal-planning dashboard, grocery list feature, and AI recipe generation

SLAM-Based Autonomous Rescue Robot 01/2025 – 05/2025
Alex To the Rescue
- Built a remotely operated rescue robot using Raspberry Pi and Arduino Mega for SLAM-based navigation, astronaut identification, and timed extractions in a lunar simulation.
- Programmed real-time control in C/C++ (Arduino) and implemented SLAM in Python (Raspberry Pi) with serial communication and multi-sensor integration under strict constraints.

EXTRACURRICULAR

NUS Raffles Hall Squash Team 03/2025 – Present
Captain
- Led weekly training sessions and represented Raffles Hall in inter-hall competitions, fostering team cohesion and performance.